SAMSUNG

AI Research Agent (RAG): LLM-powered tool for querying private document PDF sets

Team RAG

Coding and Programming Course

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Enabling People
Education for Future Generations

Project Name

About our Team

Team RAG

AI Research Agent (RAG): LLM-powered tool for querying private document

PDF sets

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Course: SIC Artificial Intelligence

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Problem Statement

Organizations and students often store important information in PDF documents such as manuals, reports, notes, and technical guides. Searching these documents manually is time-consuming and inefficient. Traditional keyword search often fails to understand context and meaning, while general AI models may generate incorrect answers if they are not grounded in source documents.

Therefore, the problem is to build a system that can intelligently read private PDF documents, retrieve the most relevant information, and generate accurate, context-based answers for user queries.

Objective of the Project

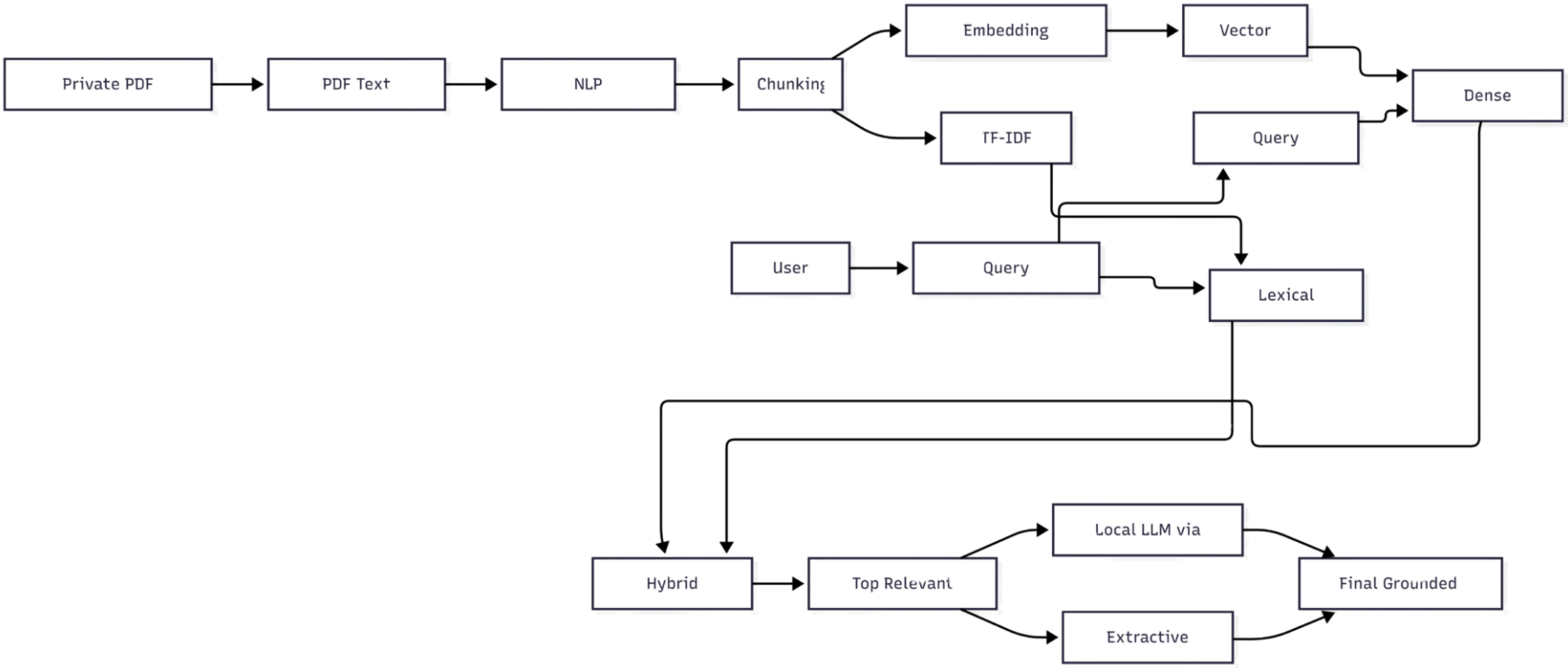
The main objective of this project is to develop an LLM-powered Retrieval-Augmented Generation (RAG) system that can answer user questions from private PDF documents. The system is designed to extract text from PDFs, preprocess and chunk the content, retrieve the most relevant information using hybrid search, and generate accurate, context-aware answers using a lightweight local LLM. The goal is to make document understanding and information retrieval faster, more intelligent, and more reliable.

Proposed Solution

Our proposed solution is an LLM-powered RAG system that reads private PDF documents, extracts and preprocesses their text, divides the content into smaller chunks, and stores them in a searchable knowledge base. When a user asks a question, the system retrieves the most relevant chunks using hybrid retrieval and then uses a lightweight local LLM to generate a grounded answer from the retrieved content.
The system combines dense semantic retrieval and lexical retrieval, which improves answer relevance for technical documents. It also supports lightweight local LLM generation through Ollama, making the solution practical for private and low-resource environments.

Unique Features -
Supports private PDF-based question answering
Uses hybrid retrieval for better relevance
Integrates a lightweight local LLM using Ollama
Falls back to extractive answering if generation is unavailable
Uses a custom Keras embedding model instead of relying only on external models

System Architecture / Flow Diagram



Tools, Technologies & Libraries

- *Project Type: AI / NLP / RAG*
- *Language: Python*
- *Libraries: NLTK, TensorFlow, Keras, NumPy, scikit-learn, PyMuPDF*
- *Tools: Jupyter, VS Code, GitHub, Ollama*
- *Dataset Source: Custom private PDF files*

Project Screenshots / Project Designs

```
PS D:\SIC> py -3.11 main.py query "What authorization methods does API Gateway support?"
WARNING:tensorflow:From C:\Users\siddh\AppData\Local\Python\pythoncore-3.11-64\Lib\site-packages\keras\src\backend\
tensorflow\core.py:232: The name tf.placeholder is deprecated. Please use tf.compat.v1.placeholder instead.

[EMBED] Loaded trained Keras encoder

============================================================
Query: What authorization methods does API Gateway support?
============================================================

API Gateway supports AWS_IAM authorization, but also allows other supported methods to control access.

Sources:
- apigateway-dg.pdf (Page 1063)
- apigateway-dg.pdf (Page 374)
- apigateway-dg.pdf (Page 1218)
- apigateway-dg.pdf (Page 47)
- apigateway-dg.pdf (Page 504)
PS D:\SIC>
```

Output

```python
def cmd_ingest(args, config):
    """Ingest PDF documents and persist processed chunks."""
    from src.ingestion.chunker import ChunkEngine
    from src.ingestion.pdf_extractor import PDFExtractor
    from src.preprocessing.nltk_processor import NLTKProcessor
    from src.retrieval.vector_store import VectorStore

    print(f"[INGEST] Scanning: {config['pdf_directory']}")

    extractor = PDFExtractor()
    documents = extractor.extract_all(config["pdf_directory"])
    print(f"[INGEST] Extracted {len(documents)} documents")

    processor = NLTKProcessor(config)
    processed_docs = []
    for doc in documents:
        processed_docs.append({**doc, "processed_text": processor.process(doc["text"])})
    print(f"[INGEST] Preprocessed {len(processed_docs)} documents")

    chunker = ChunkEngine(
        chunk_size=config["chunk_size"],
        chunk_overlap=config["chunk_overlap"],
        strategy=config.get("chunk_strategy", "sentence"),
    )
    chunks = []
```

Ingest Function

Results & Performance

Result / Final Outcome:

The project successfully developed an LLM-powered RAG system for private PDF question answering. The system can ingest PDF documents, preprocess and chunk text, retrieve relevant content using hybrid retrieval, and generate grounded answers using a lightweight local LLM through Ollama.

Key Findings:

Hybrid retrieval improved answer relevance compared to basic retrieval alone. The use of a local LLM produced more natural answers than extractive-only methods. The system worked effectively on private PDF collections and demonstrated a complete end-to-end pipeline.

Challenges Faced:

Major challenges included noisy PDF text extraction, selecting the right chunk size, retrieving relevant content from very large technical PDFs, and reducing repeated or incomplete answers. Integrating local LLM generation while keeping answers grounded was also an important challenge.

Future Scope:

The project can be improved further by adding better reranking models, support for more document types, improved citation highlighting, stronger local models, and better PDF structure understanding.

Features to be Added:

Planned features include a **frontend interface** for user interaction, drag-and-drop PDF upload, chat-style document querying, source highlighting, answer citation display, multi-document filtering, conversation history, and support for additional local LLM models.

Thank You

Open for Questions

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